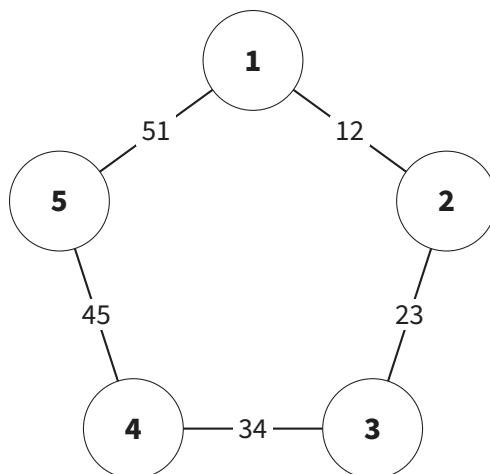


Problem 1: Start with five two-sided counters OFF. Pressing a line flips both end lamps, including ON to OFF. Write 12 for pressing the line from 1 to 2. Try two moves and undo them.



Problem 2: Start all OFF for each target. Make only the listed lamps ON. Find two different move lists, then circle your shorter list.

ON lamps	First list	Another list
1 and 2		
1 and 3		
1, 2, 3, 4		

Choose a target. Test whether one press can make it. Record what happened.

Problem 3: Start all OFF for each list. Try 12,23; then 23,12; then 12,23,12,12. Draw the final lamps for each. Cross out any pair of repeated presses that you can remove without changing the final picture.

Problem 4: A reduced list uses each edge at most once. For each reduced list below, make its target. Reset; press exactly the edges missing from that list. Compare the two final pictures and count the presses. The five edges are 12,23,34,45,51.

First list	Missing edges	Final ON lamps	Counts
12			
12,23			
12,34			
12,23,34			

Problem 5: Choose a new edge set and its missing edges. Test both. Record the two press counts and whether the final targets agree.

Problem 6: Use the five-lamp mat. To leave only lamps 1 and 3 ON, first choose whether to press 51. Lamp 1 then tells you whether to press 12. Keep going around the ring. Fill yes or no, test the full list, and check lamp 5 last.

First choice	51	12	23	34	45	Lamp 5
No	no					
Yes	yes					

Problem 7: Repeat the forced choices for only lamps 1,2,3,4 ON. Compare your two lists with your experiments on page 1.

First choice	51	12	23	34	45	Lamp 5
No	no					
Yes	yes					

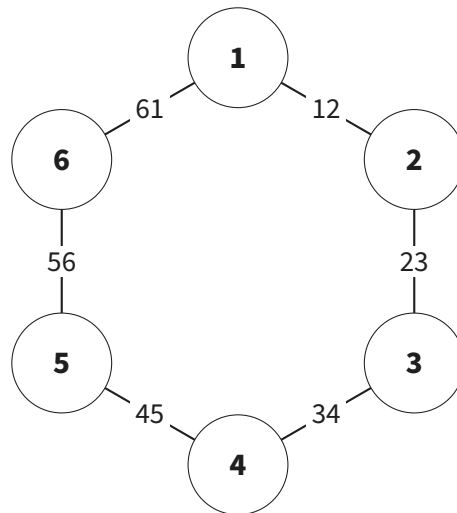
Problem 8: Now try to leave only lamp 1 ON. Fill both rows. Put a cross beside any row whose final lamp does not match the target.

First choice	51	12	23	34	45	Lamp 5
No	no					
Yes	yes					

Record a rule you notice about the two lists or their targets.

Problem 9: On the five-lamp ring, choose three new targets with an even number of ON lamps. Find the shortest reduced list for each. Record both complementary lists and their lengths. Try to find a target needing more than two presses.

Problem 10: Use six counters on this ring. Start all OFF. Make only lamps 1 and 4 ON in two different ways. Count the presses. Then repeat for only 1 and 3 ON. Compare the two lengths for each target.



Move lists, lengths, and a comparison with the five-lamp ring: